CLINICAL · EAR

Microtia and congenital canal atresia

A baby born with a small or absent ear raises two questions, and hearing comes first. Test early, amplify by six months, and only later think about building an ear or opening a canal. When surgery is planned, image first: the facial nerve is often in the wrong place.

AT A GLANCE

- Test early and amplify by 6 months.
- Thin CT before any surgery.
- Pinna is reconstructed before the canal.
- Expect an abnormal facial nerve.

1 CHAPTER ONE Overview

A first-arch malformation.

1.1 Definitions

- **Microtia:** a small, malformed pinna; *anotia* is its complete absence.
- **Congenital aural atresia:** absence or narrowing of the external canal, usually with a malformed middle ear.

The two usually occur together, on the same side.

1.2 Epidemiology

Microtia occurs in roughly 1 in 6,000 to 1 in 12,000 births, more often in boys and on the right. About 10% are bilateral.

1.3 Embryology

The pinna develops from the hillocks around the first branchial cleft, the canal from the first branchial groove, and the malleus and incus from the first and second

arches. The inner ear develops separately from the otic placode, so it is usually normal.

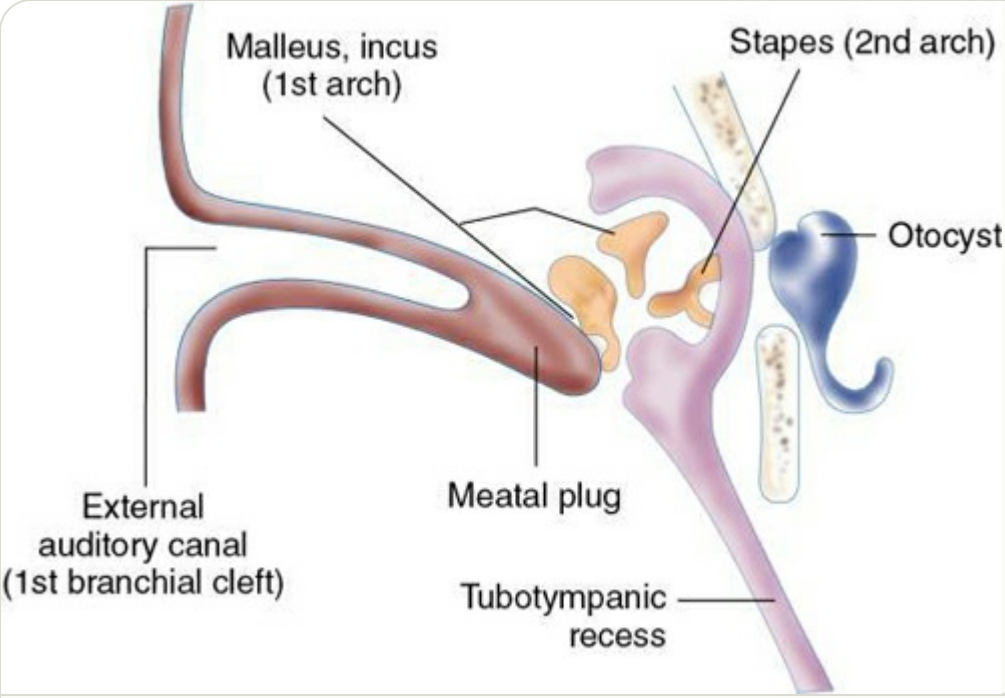


FIGURE 1 Development of the external auditory canal and middle ear.
Dhingra, fig. 1.25

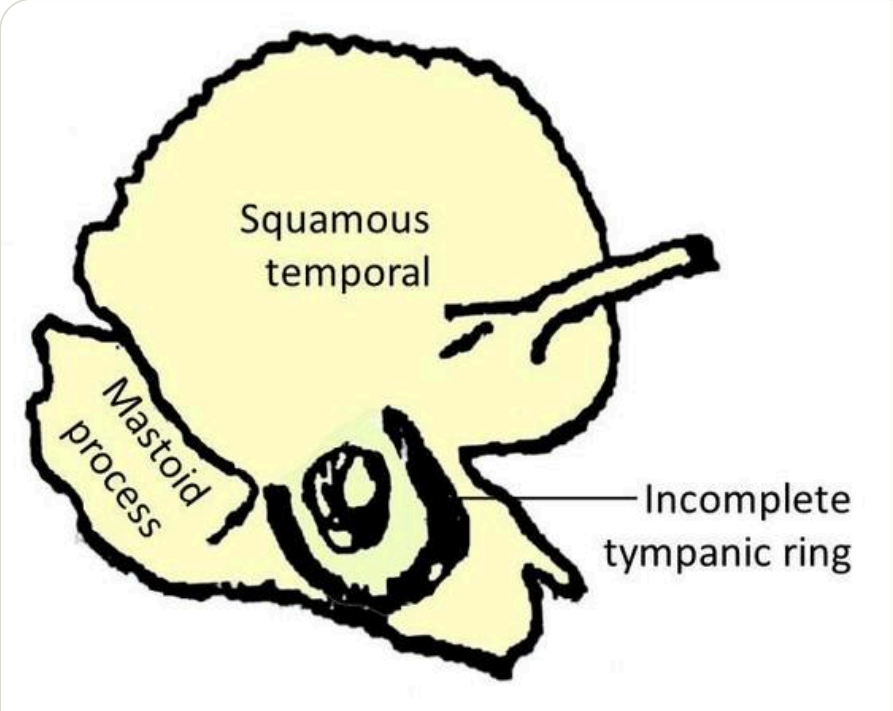


FIGURE 2 The temporal bone at birth: an incomplete tympanic ring and no mastoid process.
Anatomy atlas · Doss p. 37

1.4 Associated syndromes

Treacher Collins, Goldenhar and hemifacial microsomia, Crouzon, and branchio-oto-renal syndrome.

CHAPTER TWO

2 Classification

Pinna and middle ear.

2.1 Microtia

- **Grade I:** a recognisable but smaller or deformed pinna.
- **Grade II:** a deformed framework with some parts missing.
- **Grade III:** a soft-tissue remnant, the *peanut ear*.
- **Grade IV:** anotia.



FIGURE 3 Grade III microtia of the right ear (peanut ear).
Dhingra, fig. 8.3

2.2 Middle ear (Lee)

- **Class I:** normal pinna with an ossicular defect, the commonest.
- **Class II:** microtia and atresia with an aerated middle ear and mastoid.

- **Class III:** poorly aerated, underdeveloped middle ear and mastoid.

> **PEARL**
>
> A normal pinna with atresia implies the insult happened later in development, after the pinna had formed.

3 CHAPTER THREE Clinical features

Hearing and appearance.

3.1 Hearing

- Atresia: a conductive loss of about 60 to 70 dB.
- Ossicular malformation alone: about 25 to 50 dB.
- Congenital stapes fixation: about 50 to 65 dB.
- Bilateral atresia threatens speech and language development.

3.2 Associated findings

Preauricular tags, pits and sinuses, facial asymmetry, jaw underdevelopment, and syndromic features. Airway problems in some syndromes.

4 CHAPTER FOUR Investigations

Hearing first, imaging before surgery.

4.1 Audiology

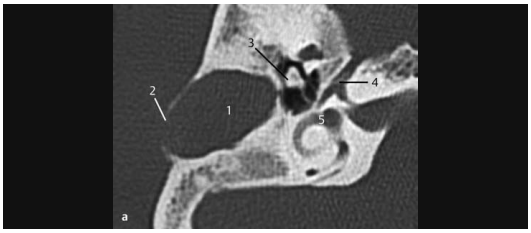
OAE and ABR from birth, including bone-conduction ABR, to measure cochlear function on each side.

4.2 Imaging

Thin-section CT of the temporal bones before any surgery, usually around five or six years: the size and aeration of the middle ear and mastoid, the ossicles, the oval and round windows, the inner ear, and the course of the facial nerve. MRI if there is a sensorineural loss.

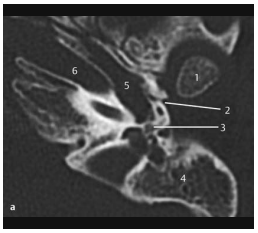
IMAGING IN THE APP 3 cases

Canal atresia with cholesteatoma, Treacher Collins syndrome, and inner ear and ossicular malformations.



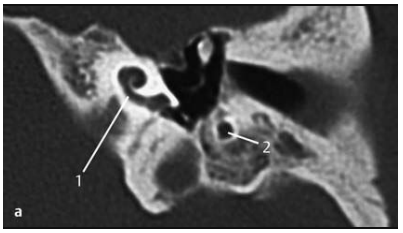
Inclusion Cholesteatoma and Atresia of the External Auditory Canal

Congenital · 3 images



Malformations, Treacher Collins

Congenital · 2 images



Malformations of the Inner Ear and Ossicular Chain

Congenital · 3 images

4.3 Other tests

Renal ultrasound when preauricular pits accompany neck fistulae (branchio-oto-renal syndrome).

5 CHAPTER FIVE Management

Hearing, then the ear.

5.1 Hearing

- **Bilateral atresia:** a bone-conduction aid on a headband within the first months, by six months at the latest.
- **Unilateral atresia with a normal other ear:** speech usually develops normally;

preferential seating at school and monitoring; amplification is optional.

- A bone-anchored implant from around five or six years.

5.2 The pinna

Reconstruction with autologous rib cartilage in stages, usually at about five to eight years when there is enough cartilage, or a porous polyethylene framework, or an osseointegrated prosthesis.

5.3 The canal

5.3.1 Selection

Canal surgery suits ears with a conductive loss over 30 dB, bone conduction better than 20 dB, and a well-aerated middle ear with a favourable facial nerve on CT. The pinna is reconstructed first so the canal can be placed correctly.

5.3.2 Risks

Facial nerve injury, restenosis of the new canal, lateralisation of the graft and sensorineural loss.

OPERATIVE GUIDES IN THEATRE 5 operations

Microtia
Microtia

Congenital and Acquired Atresia and Stenosis of the External Auditory Canal
Atresia and stenosis

Congenital Malformations of the Middle Ear
Congenital middle ear malformations

Bone-Anchored Hearing Devices
Bone-anchored hearing

Otoplasty
Otoplasty

5.4 Preauricular sinus

Leave it alone unless it becomes infected; then excise the whole tract.

THE FACIAL NERVE

In atresia the facial nerve often runs more anteriorly and laterally than usual, crossing the area where the canal would be drilled. First branchial cleft anomalies also run close

to it.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 On call

A NEWBORN WITH BILATERAL ATRESIA

1. Early audiology with bone-conduction testing.
2. A bone-conduction aid as soon as possible.
3. Look for syndromic features and airway problems.

6.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Embryology:** why the inner ear is spared.
- **Grades of microtia.**
- **Hearing before cosmetics,** and the timing of each intervention.
- **Selection for canal surgery,** and the facial nerve risk.

6.3 Common pitfalls

COMMON PITFALLS

- Delaying amplification in bilateral atresia.
- Opening a canal before reconstructing the pinna.
- Drilling without a CT map of the facial nerve.

Written for this app from Oxford Textbook of Otolaryngology ch. 13, 83, Oxford Specialist Handbook ch. 23, Dhingra ch. 8, 19 and K. J. Lee ch. 13, 17, 51. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

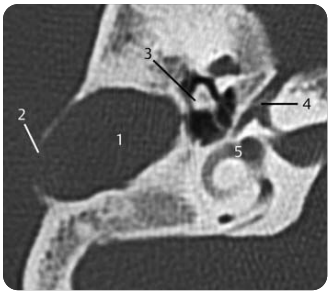
On imaging

The imaging of this topic: key findings, differentials and representative scans.

EXTERNAL EAR & TYMPANIC MEMBRANE · CONGENITAL

Inclusion Cholesteatoma and Atresia of the External Auditory Canal

- Expansion toward and/or destruction of the temporomandibular joint and toward the middle ear structures, abscess formation, and osteomyelitis and (intracranial) spread of infection.
- In case of aural atresia (first branchial groove anomaly), other dysmorphic features may be present too, especially in case of syndromal comorbidity.
- Particular attention needs to be paid to: the appearance of the middle ear cavity and mastoid pneumatization; signs of ankylosis or malformation of the ossicular chain; presence of inner ear deformities, the round and oval windows and the vestibular aqueduct; aberrations in the anterior and/or lateral course of the facial nerve, which may complicate surgery.

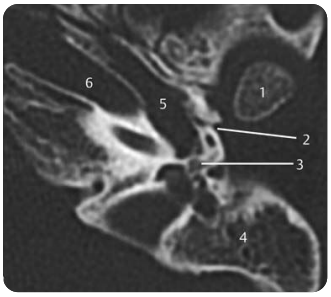


CT, axial. Expanding, round, smooth-bordered lesion (1) in the cranial part of the mastoid with partial destruction of the cortex (2). The head of the…

MIDDLE EAR & MASTOID · CONGENITAL

Malformations, Treacher Collins

- In deformities of the temporal bone, pay particular attention to the:
- External auditory canal, bony or fibrous atresia, appearance and position of the mandibular condyle and temporomandibular joint.
- Appearance of the middle ear cavity and mastoid pneumatization.
- Signs of ankylosis or disconnections of the ossicular chain.
- Presence of inner ear deformities, the round and oval windows, and the vestibular aqueduct.
- Aberrations in the course and anterior displacement of the facial nerve, which may complicate surgery.

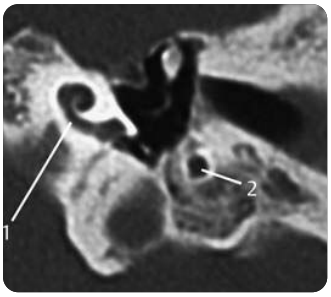


CT, axial. For orientation, this is a slice through the basal cochlear turn and the mandibular condyle (1). Behind the condyle, the bony part of the external…

INNER EAR & IAC · CONGENITAL

Malformations of the Inner Ear and Ossicular Chain

- This section illustrates the importance of a systematic comparison of both sides as well as the different compartments of the ear since bilateral congenital pathology may show (subtle) variations between the left and right side.



CT, axial. Illustrative evaluation of malformations of the inner ear as well as the middle ear. This axial slice demonstrates a cochlea consisting of…